

Planetary Rover — Single Rover, Static Environment

AI4Ro2 — Artificial Intelligence for Robotics 2 · University of Genova · 2025-26
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1. Domain and Problem Description

A single wheeled rover navigates a static directed graph of locations on a planetary surface. It starts at a base station and must collect geological samples from designated target locations, then return each sample to base. Every move between locations consumes battery, and the rover's battery capacity is strictly limited — it cannot be exceeded. The environment is fully static and fully observable: no obstacles appear or disappear, and the entire map is known before planning begins.

Objects: location, sample **Predicates:** at-rover, connected, sample-at, carrying, delivered, is-base **Numeric fluents:** battery-level, move-cost

The map used for both Q1 and Q2 is a directed graph of 4-5 locations (base, site-A, site-B, site-C, and a waypoint), with edge costs ranging from 10 to 20 units. One sample is placed at each target location, and the rover can carry only one sample at a time, forcing it to make separate round trips.

2. Modelling Choices

Q1 — Classical PDDL with numeric fluents

Battery is represented as a single global numeric fluent rather than a set of discrete propositions. The core design decision is in the move action: its precondition includes ($\geq$ battery-level move-cost), and its effect includes (decrease battery-level move-cost). This was chosen specifically because it makes energy a *genuine planning constraint* rather than a cosmetic value — the planner physically cannot select a move it cannot afford, and must therefore search for energy-feasible routes. Three actions cleanly separate the task phases: move, collect-sample, and deliver-sample.

Q2 — PDDL+ with processes, events, and continuous fluents

Q2 replaces the static, action-only battery model with three new PDDL+ constructs, chosen to capture energy behaviour that classical PDDL cannot express:

Construct	Name	Why it was used
Process	idle-drain	Battery must drain continuously, even while the rover is stationary (e.g. collecting a sample). A discrete action cannot express a continuous, always-on rate of change — only a PDDL+ process can.
Process	move-drain	An additional drain applies specifically while the rover is in motion. Modelled as a second process gated by a rover-active flag, so it runs concurrently with idle-drain rather than replacing it.
Event	battery-critical	A hard safety cutoff is needed that the planner cannot simply choose to ignore. PDDL+ events are world-triggered and fire autonomously once battery-level reaches critical-threshold, setting system-ok to false and blocking all further actions.

The two processes and the event are not explicitly linked to each other in the domain file — they coordinate purely through the shared system-ok predicate. This was a deliberate choice: it mirrors how independent physical subsystems on a real rover (power management, drive motors, safety controller) interact only through shared state, not direct calls. Drain rates are expressed per time unit (e.g. 1.0, 2.0) rather than as one-shot action costs, since processes integrate their effect over elapsed time via the #t term.

3. Plan Output Walkthrough

Q1 — Problem 1 (battery=60): ENHSP returns a 4-step plan — move, collect, move, deliver — consuming 20 of the 60 available battery units. This is the expected trivial case: battery is comfortable and the planner finds the

direct round trip with no search difficulty (6 expanded nodes).

Q1 — Problem 2 (battery=130, 3 samples): ENHSP returns a 16-step plan covering three separate round trips. For the third sample at site-C, the planner correctly routes via the waypoint (base→waypoint1→site-C, cost 35) rather than via site-A and site-B (cost 10+10+15=35) — both paths cost the same, and the planner selects the one requiring fewer intermediate actions. The plan consumes exactly 130 battery units, confirming the budget is tight but sufficient. This matches the intended design: changing only the initial battery value forces a structurally different, longer plan compared to Problem 1.

Q2 — Problem 1 (battery=80): the grounded problem now reports $|P|=1$ $|E|=1$, confirming the idle-drain process and battery-critical event are active in the search. ENHSP (a satisficing, discrete-event planner) reports all four actions at elapsed time 0, since it finds the shortest valid action sequence under its time-stepping model; the important confirmation is that the plan remains valid throughout, i.e. system-ok is never violated before the goal is reached.

Q2 — Problem 2 (battery=80, 2 samples): an 8-step plan is found, again with $|P|=1$ $|E|=1$. The planner chooses to visit site-B first (direct edge, cost 20) and site-A second (cost 10 there and back via base), rather than the reverse order — both are valid, and the planner's choice satisfies the goal while keeping the final battery level above the critical-threshold of 5. This confirms that the PDDL+ model behaves as expected: the event remains armed but does not fire, because the chosen plan stays within the energy budget.

All four plans were verified by manually tracing battery levels against each move-cost and drain rate; the planner's reported metrics match these manual calculations exactly.

4. Limitations and Known Issues

- **Instantaneous moves.** Travel between locations takes zero simulated time. ENHSP in this configuration does not support durative actions for PDDL+ problems, so true travel time could not be modelled.
- **No geometry.** The connected predicate assumes a traversable path exists; no terrain, obstacles, or path planning are represented.
- **No recharge.** Battery is strictly decreasing in both models — no solar or RTG charging action or process is included.
- **Single sample capacity.** The rover can carry only one sample at a time; this is an implicit consequence of the action structure rather than an explicit capacity fluent.
- **No uncertainty.** Both models are fully deterministic. Real rovers face sensor noise, actuator slip, and communication delay, which would require an MDP/POMDP formulation rather than PDDL/PDDL+.
- **Linear drain only.** Battery degradation in Q2 is linear in time; real battery behaviour is non-linear (temperature, age, discharge curve effects).
- **Scalability of the PDDL+ model.** An earlier, larger version of Problem 2 (5 locations, battery=310) caused ENHSP to exhaust available heap memory during search; the problem was simplified to 3 locations and battery=80 to remain solvable, without changing the underlying lesson about timing-dependent feasibility.

5. Output

Full planner output (raw ENHSP logs) for all four problem instances is included in the repository at codes/output/Q1-output.txt and codes/output/Q2-output.txt, including grounding statistics, search expansion counts, and the complete found plans reproduced above.